

Course COS 332 S1 2020
Test Semester test 2
Started 6/23/20 5:27 PM
Submitted 6/23/20 7:10 PM
Due Date 6/23/20 7:30 PM
Status Completed
Attempt Score 31 out of 50 points
Time Elapsed 1 hour, 43 minutes out of 2 hours

- Instructions
- COS332 Semester Test 2 (23 June 2020)
 - This is a two hour test to be written at any time in the window from 15:30 to 19:30.
 - Note that the time allocated is longer than what would typically have been allocated, given the challenges of writing the test in a context where you may encounter connectivity problems and other technical challenges.
 - This is a closed-book test. You are not allowed to obtain help from any person or other source while writing the test. You are, however, allowed to email the lecturer during the test at molivier@cs.up.ac.za if anything is unclear.
 - You have two hours to complete the test after you started writing. However, all writing will be suspended at 19:30, even if you started writing less than two hours before that time.

Results Submitted Answers, Correct Answers
Displayed

• **Question 1**

0 out of 0 points

Do you promise not to consult any prohibited resources while writing this test?

Selected Answer:  True

Correct Answer:  True

• **Question 2**

0 out of 1 points

For this and other multiple choice questions in the paper, select the alternative that fits the question best in each case.

The routing prefix in an IPv6 address (normally) consists of the ... of the address.

Selected Answer:  First 32 bits

Correct Answer:  First 64 bits

• **Question 3**

0 out of 1 points

A token is reserved as follows on a token ring: The node wishing to reserve the token

Selected Answer: 

Simply increases its priority.

Correct Answer: 

None of the above

- **Question 4**

0 out of 1 points

To which ISO OSI layer does TCP/IP's Internet layer map?

Selected Answer: 

2

Correct Answer: 

3

- **Question 5**

0 out of 1 points

Advantages of a LEO satellite for data communications include the following:

Selected Answer: 

More than one of the above

Correct Answer: 

Since it is placed closer to earth than geosynchronous satellites, a smaller transmitter can be used from earth.

- **Question 6**

1 out of 1 points

Most layer 3 protocols accomplish routing by

Selected Answer: 

Consulting routing tables compiled by layer 7 routing applications.

Correct Answer: 

Consulting routing tables compiled by layer 7 routing applications.

- **Question 7**

1 out of 1 points

What is the purpose of the netmask?

Selected Answer: 

Used by routers to identify the network portion from the host portion of an IP address to enable proper routing of packets

Correct Answer: 

Used by routers to identify the network portion from the host portion of an IP address to enable proper routing of packets

- **Question 8**

1 out of 1 points

IP fragmentation occurs because of

Selected Answer: 

Different MTUs on the Internet.

Correct Answer: 
Different MTUs on the Internet.

• **Question 9**

1 out of 1 points

Which of the following is not a valid DNS record type?

Selected Answer: 
AA

Correct Answer: 
AA

• **Question 10**

0 out of 1 points

Which of the following URLs is/are valid?

Selected Answer: 
<http://www.cs.up.ac.za/>

Correct Answer: 
More than one of the above

• **Question 11**

0 out of 1 points

Which IEEE standard specifies the operation of wireless local area networks?

Selected Answer: 
IEEE 802.3

Correct Answer: 
IEEE 802.11

• **Question 12**

1 out of 1 points

Layer 6 in the ISO OSI protocol stack is known as the ___ layer.

Selected Answer: 
Presentation

Correct Answer: 
Presentation

• **Question 13**

1 out of 1 points

The entry 0.0.0.0/0 in a routing table is used to provide information about the

Selected Answer: 
Default gateway.

Correct Answer: 
Default gateway.

• **Question 14**

1 out of 1 points

Which of the following is not a valid second-level domain in the za ccTLD?

Selected Answer: 
com

Correct Answer: 
com

- **Question 15**

1 out of 1 points

UMTS has been designed to be a logical upgrade path for

Selected Answer: 
GSM

Correct Answer: 
GSM

- **Question 16**

1 out of 1 points

IP datagrams support source routing. Source routing is

Selected Answer: 
Where the source determines the route of the packet.

Correct Answer: 
Where the source determines the route of the packet.

- **Question 17**

1 out of 1 points

If parity is used for error control, only errors that affect ... will be detected.

Selected Answer: 
An odd number of bits

Correct Answer: 
An odd number of bits

- **Question 18**

1 out of 1 points

Hop count refers to

Selected Answer: 
A routing metric.

Correct Answer: 
A routing metric.

- **Question 19**

0 out of 1 points

X.509 is a standard of the

Selected Answer: 
IEEE

Correct Answer: 
ITU

- **Question 20**

1 out of 1 points

A modem

Selected Answer: 
Answer:

Modulates a digital signal so that it can be transmitted via an analog medium over a long distance.

Correct
Answer:



Modulates a digital signal so that it can be transmitted via an analog medium over a long distance.

- **Question 21**

1 out of 1 points

Indicating where layer 2 frames start and end is known as

Selected Answer:



Data delineation

Correct Answer:



Data delineation

- **Question 22**

0 out of 1 points

An ARP cache is maintained on

Selected Answer:



Layer 4 of the protocol stack.

Correct Answer:



Each host connected to an Ethernet bus.

- **Question 23**

1 out of 1 points

What is the name of ISO layer 2?

Selected Answer:



Data link layer

Correct Answer:



Data link layer

- **Question 24**

1 out of 1 points

Where are fragmented IP datagrams reassembled?

Selected Answer:



At the final destination.

Correct Answer:



At the final destination.

- **Question 25**

1 out of 1 points

How long in bits is an IPv6 address?

Selected Answer:



128

Correct Answer:



128

- **Question 26**

0 out of 1 points

An IPv4 multicast packet

Selected Answer: 

Should be delivered to all subscribed parties.

Correct Answer: 

More than one of the above

- **Question 27**

0 out of 1 points

If the International Standards Organisation (ISO) adopts a standard

Selected Answer: 

All member countries have to comply with the standard, but the rest of the world do not have to.

Correct Answer: 

None of the above

- **Question 28**

0 out of 1 points

Which of the following IP addresses does not belong to a private IP address block?

Selected Answer: 

10.11.12.13

Correct Answer: 

238.239.240.241

- **Question 29**

1 out of 1 points

The ASCII STX character is an abbreviation for

Selected Answer: 

Start of text

Correct Answer: 

Start of text

- **Question 30**

0 out of 1 points

IP is deemed to be a reliable protocol (in the same sense as TCP) because

Selected Answer: 

IP datagrams are acknowledged upon receipt.

Correct Answer: 

IP is, in fact, deemed to be unreliable.

- **Question 31**

0 out of 1 points

MACA is an example of

Selected Answer: 

Packet passing

Correct Answer: 

CSMA/CA

- **Question 32**

0 out of 1 points

Consider data delineation as implemented by various data link protocols. Answer this and the following four questions in such a context. Note that answers are NOT case sensitive.

If the DATA

011111010

is to be transmitted via SDLC, how will this data (with no added surrounding 'markers') be transmitted?

Selected Answer: ❌ 011111001111101001111110

Correct Answer:

Evaluation Method	Correct Answer	Case Sensitivity
✅ Exact Match	0111110010	

- **Question 33**

1 out of 1 points

The bit pattern 01111110 used in SDLC/HDLC to delineate frames is known as a ____.

Selected Answer: ✅ flag

Correct Answer:

Evaluation Method	Correct Answer	Case Sensitivity
✅ Exact Match	Flag	

- **Question 34**

1 out of 1 points

The fact that a BSC message cannot, for example, transmit a stream of octets that happen to contain a bit pattern that looks like the EOT character encoded in ASCII is known as the ____ problem.

Selected Answer: ✅ transparency

Correct Answer:

Evaluation Method	Correct Answer	Case Sensitivity
✅ Exact Match	Transparency	
✅ Exact Match	Transparancy	

- **Question 35**

1 out of 1 points

The start of frame delimiter follows the preamble in an Ethernet frame. Use a sequence of zeroes and ones to indicate the value of this start of frame delimiter. Do not insert any spaces into your answer.

Selected Answer: ✅ 10101011

Correct Answer:

Evaluation Method	Correct Answer	Case Sensitivity
✅ Exact Match	10101011	

- **Question 36**

1 out of 1 points

The payload of an Ethernet frame cannot exceed ___ octets.

Selected Answer:  1500

Correct Answer:  1,500

Answer range +/- 0 (1500 - 1500)

- **Question 37**

1 out of 1 points

This question tests whether you understand Hamming code as it has been explained in class.

You use a Hamming code capable of correcting single bit errors. Bits are numbered from right to left. Even parity is used. You receive the character below.

Indicate in the appropriate space which bit has changed by only writing a number from 1 to 12. If no bit has changed, indicate it by writing 0.

100101101011

Selected Answer:  3

Correct Answer:  3

Answer range +/- 0 (3 - 3)

- **Question 38**

1 out of 1 points

This question tests whether you understand Hamming code as it has been explained in class.

You use a Hamming code capable of correcting single bit errors. Bits are numbered from right to left. Even parity is used. You receive the character below.

Indicate in the appropriate space which bit has changed by only writing a number from 1 to 12. If no bit has changed, indicate it by writing 0.

000100011110

Selected Answer:  9

Correct Answer:  9

Answer range +/- 0 (9 - 9)

- **Question 39**

1 out of 1 points

This question tests whether you understand Hamming code as it has been explained in class.

You use a Hamming code capable of correcting single bit errors. Bits are numbered from right to left. Even parity is used. You receive the character below.

Indicate in the appropriate space which bit has changed by only writing a number from 1 to 12. If no bit has changed, indicate it by writing 0.

000110110001

Selected Answer:  3

Correct Answer:  3

Answer range +/- 0 (3 - 3)

- **Question 40**

1 out of 1 points

This question tests whether you understand Hamming code as it has been explained in class.

You use a Hamming code capable of correcting single bit errors. Bits are numbered from right to left. Even parity is used. You receive the character below.

Indicate in the appropriate space which bit has changed by only writing a number from 1 to 12. If no bit has changed, indicate it by writing 0.

110000101010

Selected Answer:  7

Correct Answer:  7

Answer range +/- 0 (7 - 7)

- **Question 41**

1 out of 1 points

This question tests whether you understand Hamming code as it has been explained in class.

You use a Hamming code capable of correcting single bit errors. Bits are numbered from right to left. Even parity is used. You receive the character below.

Indicate in the appropriate space which bit has changed by only writing a number from 1 to 12. If no bit has changed, indicate it by writing 0.

010101011101

Selected Answer:  6

Correct Answer:  6

Answer range +/- 0 (6 - 6)

- **Question 42**

1 out of 1 points

Assume 133.133.0.0 is a class B IPv4 address that is used by your company. You are given the task to subnet it such that at least 100 subnets are supported, with support for at least 500 computers on each subnet. Use this information for this question, as well as the four questions that follow.

What netmask will your company use internally after the subnetting?

Selected Answer:  255.255.254.0

Correct Answer:

Evaluation Method	Correct Answer	Case Sensitivity
 Exact Match	255.255.254.0	

- **Question 43**

0 out of 1 points

Consider subnet 40. The its network address will be of the form 133.133.x.y/z.

What is the value of x?

Selected Answer:  40

Correct Answer:  80

Answer range +/- 0 (80 - 80)

- **Question 44**

1 out of 1 points

Consider subnet 40. The its broadcast address will be of the form 133.133.v.w

What is the value of w?

Selected Answer:  255

Correct Answer:  255

Answer range +/- 0 (255 - 255)

- **Question 45**

0 out of 1 points

Consider subnet 40. The its broadcast address will be of the form 133.133.v.w

What is the value of v?

Selected Answer:  41

Correct Answer:

Evaluation Method	Correct Answer	Case Sensitivity
 Exact Match	81	

- **Question 46**

0 out of 1 points

Consider the address 133.133.133.133. On which subnet will it be located?

Selected Answer:  132

Correct Answer:  66

Answer range +/- 0 (66 - 66)

- **Question 47**

1 out of 1 points

In the video on IPv6 I stated that the localhost address is "all zeroes (::)". This was wrong. The localhost address is the number 1 (in decimal, binary, hexadecimal, octal and the vast majority of number systems you ought to care about). So, what is the correct localhost address, expressed in colon-hexadecimal notation with zero suppression?

(Hint: It is NOT :: as I said in the video.)

Selected Answer:  ::1

Correct Answer:

Evaluation Method	Correct Answer	Case Sensitivity
 Exact Match	::1	

- **Question 48**

0 out of 1 points

The command

ping ::ffff.1.2.3.4

will be handled by the ... at the origin site.

Selected Answer: 

IPv6 and IPv4 stacks

Correct Answer: 

IPv4 stack

- **Question 49**

1 out of 1 points

The host number of an IPv6 address consists of ___ bits.

Selected Answer:  64

Correct Answer:  64

Answer range +/- 0 (64 - 64)

- **Question 50**

1 out of 1 points

Which prefix is used to indicate an IPv6 autoconfigured / link-local address?

Selected Answer:  fe80::/10

Correct Answer:

Evaluation Method	Correct Answer	Case Sensitivity
 Exact Match	fe80::/10	

- **Question 51**

0 out of 1 points

An IPv6 routing prefix consists of ___ bits.

Selected Answer:  96

Correct Answer:  64

Answer range +/- 0 (64 - 64)

- **Question 52**

0 out of 0 points

Have you cheated while writing this test?

Selected Answer: 
False

Correct Answer:  False

Sunday, April 11, 2021 6:36:13 PM SAST